

Test Plan and Report

Product: Nodegent
Date: June 08, 2026

Team: Nodegent
App URL: <https://nodegent.app>

System Test Scenarios

Each scenario corresponds to one or more user stories and their acceptance criteria. Scenarios were run against the released system at <https://nodegent.app>.

| Scenario 1: Sign-In with Google (PASS)

User Story US-2.1: *As a student, I want to sign in using my campus or Google credentials so my experience is personalized and secure.*

1. Navigate to <https://nodegent.vercel.app>.
→ *Expected: Landing page loads with a visible 'Sign In' button.*
2. Click 'Sign In'.
→ *Expected: Redirected to /sign-in; Clerk sign-in widget is visible.*
3. Click 'Continue with Google'; select a Google account (e.g. test@ucsc.edu).
→ *Expected: OAuth consent screen appears. After approval, user is redirected to /dashboard.*
4. Refresh the page at /dashboard.
→ *Expected: Session persists; user remains on /dashboard without re-authenticating.*
5. Open a new incognito tab and navigate to <https://nodegent.vercel.app/dashboard>.
→ *Expected: Unauthenticated user is immediately redirected to /sign-in.*

| Scenario 2: Canvas Authentication & Course Sync (PASS)

User Story US-2.2: *As a student, I want Nodegent to read my enrolled courses and upcoming assignments from Canvas.*

6. Sign in to Nodegent (see Scenario 1). Navigate to the Campus Sync panel.
→ *Expected: Canvas sync section shows status 'Not Connected'.*
7. Click 'Connect Canvas'. Enter CruzID and Gold Password in the credential form and submit.
→ *Expected: A live browser mirror appears; Duo MFA prompt is shown.*
8. Approve the Duo MFA push on your registered device.
→ *Expected: Browser mirror completes authentication. A success message appears: 'Canvas connected'.*
9. Wait for the initial sync to complete (progress indicator visible).
→ *Expected: Dashboard populates with enrolled courses and upcoming assignments from Canvas.*
10. Open DevTools → Network tab. Verify no response body contains 'canvasCookies' or raw session tokens.

→ Expected: No cookies or credentials appear in any client-accessible response.

| Scenario 3: Assignment Dashboard & Completion (PASS)

User Story US-2.3: As a student, I want a dashboard showing all my upcoming assignments in priority order.

11. After Canvas sync (Scenario 2), navigate to the main dashboard.
→ Expected: Assignment list is visible with titles, course names, and due dates.
12. Verify the sort order: the assignment with the nearest due date appears first.
→ Expected: Assignments are sorted ascending by due date; overdue items show a red badge.
13. Click the checkbox on the topmost assignment to mark it complete.
→ Expected: Assignment disappears from the active list immediately.
14. Click 'Show completed' toggle at the top of the assignment list.
→ Expected: The just-completed assignment reappears with a strikethrough or 'Completed' indicator.

| Scenario 4: AI Chat Assistant (PASS)

User Story US-3.1: As a student, I want to chat with an AI assistant that already knows my courses, assignments, and schedule.

15. From the dashboard, open the AI Chat panel (click the chat icon in the dock or press Cmd+K).
→ Expected: Chat interface opens with an empty message input.
16.
→ Expected: AI responds within a few seconds with a list of this week's assignments sourced from the user's Canvas data — not generic placeholder text.
17.
→ Expected: AI identifies upcoming exams or quizzes from synced Canvas data and states dates accurately.
18. Open DevTools → Network tab and inspect the chat API request.
→ Expected: No OpenAI or Anthropic API key appears in the request payload or response — LLM calls are server-side only.
19. After the conversation, navigate to the Activity Log.
→ Expected: Two AI chat entries appear in the log, each with a timestamp.

| Scenario 5: Google Calendar Sync (PASS)

User Story US-3.2: As a student, I want to connect my Google Calendar and have Nodegent automatically create events for my deadlines and exams.

20. Navigate to Campus Sync → Google Calendar section. Click 'Connect Google Calendar'.
→ Expected: Google OAuth consent screen opens within the app. After approval, status changes to 'Connected'.
21. Click 'Sync Now'.
→ Expected: Progress indicator appears briefly, then status updates to 'Last synced: just now'.
22. Open Google Calendar in a browser tab.

→ Expected: Assignment due dates from Canvas appear as calendar events. Exams/quizzes are present as all-day or timed events.

23. Return to Nodegent and click 'Sync Now' again.

→ Expected: No duplicate calendar events are created; existing events are updated if details changed.

24. Navigate to the Activity Log.

→ Expected: Two 'calendar_sync' entries appear with timestamps and 'success' status.

Scenario 6: Access Toggles (PASS)

User Story US-3.4: As a student, I want to choose which systems Nodegent can access using simple on/off toggles.

25. Navigate to Settings → Access Controls.

→ Expected: Canvas and Google Calendar toggles are visible and currently ON.

26. Toggle Canvas OFF.

→ Expected: Toggle state saves immediately. A confirmation or status indicator reflects 'Canvas: Disabled'.

27.

→ Expected: AI responds without any course or assignment data — Canvas context is excluded from the AI prompt.

28. Toggle Canvas back ON.

→ Expected: Toggle reflects enabled state. Next AI query can reference Canvas data again.

Scenario 7: Activity Log (PASS)

User Story US-4.1: As a student, I want to see a clear activity log of everything Nodegent has done on my behalf.

29. After completing Scenarios 2–6, navigate to the Activity Log from the dock or Settings.

→ Expected: Log is visible and accessible without extra navigation.

30. Review the log entries.

→ Expected: Entries exist for: Canvas sync (with timestamp), AI chat messages (with timestamp), Calendar sync (with timestamp), Access toggle changes.

31. Verify entries are sorted newest-first.

→ Expected: Most recent action appears at the top of the list.

32. Sign in as a different user in a separate browser. Navigate to their Activity Log.

→ Expected: The second user sees only their own log entries — none from the first user.

Scenario 8: Revoke Canvas Access (PASS)

User Story US-4.2: As a student, I want to pause or revoke Nodegent's access to any connected system instantly.

33. Navigate to Settings → Security & Access. Click 'Disconnect Canvas'.

→ Expected: Confirmation prompt appears. After confirming: Canvas status changes to 'Not Connected'; all synced courses and assignments are removed from the dashboard immediately.

34. Attempt to trigger a Canvas sync (e.g. via the Campus Sync panel).
→ *Expected: Sync does not run. UI reflects that Canvas is not connected.*
35. Navigate to the Activity Log.
→ *Expected: A 'canvas_revoke' or equivalent entry appears with a timestamp recording the revocation.*
36. Navigate to Settings → Security & Access. Click 'Disconnect Google Calendar'.
→ *Expected: Calendar status changes to 'Not Connected'; calendar events sourced from Google Calendar are removed from the events view. Assignment googleCalendarEventId fields are cleared.*

Unit Tests

Automated tests were written using Vitest (unit/integration) and Playwright (end-to-end). All tests pass against the released system version. Be prepared to run during the project review.

Vitest — Convex Backend Unit Tests

Directory: convex/___tests___/

Run with: npm run test

Test File	Coverage	Status
canvas.test.ts	getCanvasStatus (auth, isolation, cookie exposure, needsReconnect lifecycle), removeCanvasCredentials, syncCanvas 401/500 handling	PASS
assignments.test.ts	getUpcomingAssignments (sort, filter, no-due-date), upsertAssignment dedup, user isolation, markComplete auth, course summary denormalization	PASS
auditLog.test.ts	logCalendarSync entry creation, newest-first ordering, cross-user isolation, access_toggle audit entry, unauthenticated no-op	PASS
accessToggles.test.ts	updateAccessToggles (canvasEnabled, calendarEnabled), getUserSettings, AI context exclusion when toggle is off, default-enabled for new users	PASS
revocation.test.ts	revokeCanvasAccess (deletes credentials, courses, assignments), revokeCalendarAccess (deletes events, clears sync fields, clears googleCalendarEventId)	PASS

Test File	Coverage	Status
chat.test.ts	sendMessage auth guard, message creation, campus-aware context injection, cross-user thread isolation	PASS
googleCalendar.test.ts	getCalendarSyncStatus, updateCalendarSyncStatus, calendar token management	PASS
notifications.test.ts	isNew flag on first insert vs. re-sync, cross-user notification isolation	PASS
streak.test.ts	streak increment on completion, reset on missed day, longestStreak tracking	PASS
streak.helpers.test.ts	Pure helper function unit tests for streak date math	PASS
courses.test.ts	upsertCourse dedup, getCourses isolation, recomputeCourseSummaryPublic auth	PASS
courseNotes.test.ts	Note creation, update on blur, per-user/per-course isolation	PASS
users.test.ts	ensureUser idempotency, getCurrentUser, updateAccessToggles persistence	PASS
events.test.ts	Event insert/query, source filtering, user isolation	PASS

Vitest — Frontend Unit Tests

Directory: src/

Test File	Coverage	Status
src/app/dashboard/components/__tests__/AssignmentCard.test.tsx	Renders assignment title, due date, urgency badge; completion toggle interaction	PASS
src/app/dashboard/components/courses/__tests__/courseNotes.helpers.test.ts	Auto-save debounce logic, empty-state handling	PASS
src/lib/__tests__/google-calendar.test.ts	Google Calendar event construction, deduplication key generation	PASS

Playwright — End-to-End Tests

Directory: e2e/

Run with: npm run test:e2e

Test File / Scenario	Coverage	Status
sign-in.spec.ts — Unauthenticated redirect	Visiting /dashboard without auth redirects to /sign-in	PASS
sign-in.spec.ts — Clerk widget visible	Sign-in page renders Clerk component	PASS
sign-in.spec.ts — Landing page Sign In link	Landing page shows Sign In link pointing to /sign-in	PASS
sign-in.spec.ts — Landing page heading	Landing page displays Nodegent heading	PASS
snapshot-digest.spec.ts — Unauthenticated redirect	Dashboard redirects unauthenticated users; snapshot button is unreachable	PASS
snapshot-digest.spec.ts — Authenticated digest tests (3 tests)	Weekly digest banner, dismiss behavior, study timer. Require saved Clerk auth state (e2e/.auth/user.json).	SKIPPED

Note: The 3 skipped Playwright tests require a saved Clerk authentication state file (e2e/.auth/user.json), which is generated by running the sign-in automation flow. These tests are marked test.skip in the source and are ready to enable once the auth state file is in place. All non-skipped tests pass.